

Zibo Liu

Full Public Academic Curriculum Vitae

Permanent Researcher, Department of Engineering Mechanics, KTH Royal Institute of Technology
zibo@kth.se · KTH profile · ORCID 0000-0002-7547-6535 · Google Scholar · Scopus 57202216324

RESEARCH PROFILE

Zibo Liu is an acoustics researcher working across physical acoustics, acoustic metamaterials, computational design, and human-centred soundscape engineering. He combines wave-physics modelling, numerical and data-driven methods, experiments, and perceptual evaluation. He is Principal Investigator of the Swedish Research Council-funded SMILE project at KTH.

13 journal articles	475 citations	9 h-index	9 i10-index
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Public version. This comprehensive academic CV omits private contact details, confidential project material, and unpublished intellectual property. Bibliometric data are maintained from the canonical CV source.

APPOINTMENTS AND EXPERIENCE

Apr 2026–present	Permanent Researcher (20%) , Marcus Wallenberg Laboratory for Sound and Vibration Research, Department of Engineering Mechanics, KTH Royal Institute of Technology. Principal Investigator of SMILE.
Jan 2025–Mar 2026	Affiliated Researcher , KTH Royal Institute of Technology. Developed collaborative research in soundscape engineering, acoustic metamaterials, and structure-borne sound.
Jan 2023–present	Senior Acoustic Research Engineer (promoted Oct 2025; previously Acoustic Research Engineer), commercial-vehicle research and development, Stockholm. Work includes structural dynamics, NVH simulation, sound design, and academic supervision.
Jan 2024–Jan 2025	Affiliate Academic , Acoustic Research Group, Institute for Environmental Design and Engineering, University College London. Soundscape research for engineered environments.
May 2021–Apr 2023	Postdoctoral Researcher , State Key Laboratory of Tribology in Advanced Equipment, Department of Mechanical Engineering, Tsinghua University. Principal Investigator of projects on phonons, friction, and superlattices; co-supervision and teaching.
Jul 2022–Mar 2023	Visiting Postdoctoral Researcher , KTH Royal Institute of Technology. Research on acoustic metamaterials, superlattices, and railway noise and vibration.
Apr 2019–Apr 2021	Assistant Researcher , National Innovation Institute for Defense Technology, Academy of Military Sciences, Beijing. Project management and modelling for thermal and acoustic insulation.

EDUCATION

2014–2019	PhD in Engineering Acoustics , KTH Royal Institute of Technology. Thesis: <i>Design of Soundproof Panels via Metamaterial Concept</i> . Supervisors: Leping Feng and Romain Rumpler.
2011–2014	MSc in Theoretical Acoustics , National University of Defense Technology. Thesis: <i>Acoustic Lens Design for Sound Focusing</i> . Supervisor: Xinwu Zeng.
2007–2011	BEng in Applied Mechanics with a minor in Business Administration, Beijing Institute of Technology.

RESEARCH GRANTS AND FELLOWSHIPS

2026–2029	SMILE: Soundscape engineering with Metamaterial Innovation and LEarning Principal Investigator · Swedish Research Council · Grant 2025-06253 · SEK 3,500,000.
2025–2026	SMILE City: A Pre-study Co-Principal Investigator · Stockholm Trio–UCL Collaborative Research Seed Fund · KTH–UCL collaboration in acoustic metamaterials and soundscape research.
2022–2024	Moiré phonon contribution to energy dissipation in nanofriction Principal Investigator · National Natural Science Foundation of China · Grant 52205209.
2022–2024	Superlubrication enabled by superlattices for low-dimensional materials Principal Investigator · China Postdoctoral Science Foundation · Grant 2022M711805.
2021	Postdoctoral Fellowship , Office of China Postdoc Council, hosted by Tsinghua University.
2014	Doctoral Fellowship , China Scholarship Council, hosted by KTH Royal Institute of Technology.

SUPERVISION

Doctoral research

- **Xiang Fang**, University College London (2024–present), soundscape and acoustic metamaterials — co-supervisor.
- **Haolei Dai**, Tsinghua University (2021–2023), phonon contributions to nanofriction — co-supervisor.
- **Yujin Wang**, Tsinghua University (2021–2023), phonon engineering and superlattices — co-supervisor.

Master's research

- **Vivek Rajkumar Anbazhagan**, Linköping University (2024–2025), acoustic simulation for silencer design — main supervisor and project lead.
- **Merna Danka and Hannes Salomonsson**, KTH (2024–2025), soundscape approach for cabin acoustics — main supervisor and project lead.
- **Chenhao Bi**, University College London (2024–2025), soundscape approach for cabin acoustics — co-supervisor.
- **Lujie Xu**, Tsinghua joint programme (2020–2021), phononic crystals for energy-dissipation control — co-supervisor.

TEACHING AND PEDAGOGY

2024–2026	Teaching Assistant and section lead , BENV0033 Building Acoustics, University College London.
2022, 2023	Guest Lecturer , Nanomanufacturing and Interface Science, Tsinghua University. Graduate lectures on phonon properties and engineering.
2021–2023	Research seminar leader , weekly Phonons and Phononic Crystals focus group, Tsinghua University.
2022	Research Supervision and Mentoring , 16-hour pedagogical course, Tsinghua University.
2016	Basic Communication and Teaching , 20-hour pedagogical course, KTH.

ACADEMIC SERVICE AND LEADERSHIP

Peer review

Regular reviewer since 2018 for journals including *Mechanical Systems and Signal Processing*, *International Journal of Mechanical Sciences*, *Journal of Sound and Vibration*, *Thin-Walled Structures*, *Applied Acoustics*, and *JASA Express Letters*.

Conference service

Session chair, “Metamaterials for Noise Control,” 27th International Congress on Sound and Vibration, 2021.

Research leadership

Principal Investigator and project leader for SMILE; founder and leader of a weekly phonon-engineering research forum at Tsinghua; project manager for multi-partner acoustic simulation and sound-design activities.

Research environment

Laboratory equipment committee member and research-safety protocol contributor at Tsinghua; contributor to the European Shift2Rail FINE-2 project at KTH.

Mentorship

Research mentoring for early-career researchers and graduate students, including support for research planning, scientific writing, and publication strategy.

SELECTED INVITED AND CONFERENCE PRESENTATIONS

Jul 2024	An engineering framework towards a better soundscape in truck cabins using metamaterial absorber , ICSV 30, Amsterdam.
Jul 2021	Design of curved sandwich panels to overcome ring-frequency and coincidence effects , ICSV 27.
Jun 2021	Curved double wall with embedded resonators to improve sound transmission loss , ICCS 24, University of Porto.
Jul 2018	Acoustic behaviour of a locally resonant metamaterial curved panel , ICSV 25, Hiroshima.
Jun 2017	Broadband acoustic metamaterials for low-frequency noise insulation , Acoustics '17, Boston.
Jun 2016	Finite-element vibro-acoustic analysis of plates and sandwich structures , BNAM 2016, Stockholm.

GRANTED PATENTS

1. **CN115343269B** (granted 2023). Method and system for regulating material thermal conductivity through phonon-defect engineering. Co-inventors: Dameng Liu, Yujin Wang, Haolei Dai, and Zibo Liu.
2. **CN214895044U** (granted 2021). Test rig for evaluating sound insulation performance of pipeline insulation structures. Co-inventor.
3. **CN112834624B** (granted 2024). Test bench and method for evaluating sound insulation performance of pipeline insulation structures. Co-inventor.

PROFESSIONAL MEMBERSHIPS AND RECOGNITION

- Member, Acoustical Society of America (2016–present).
- Member, International Institute of Acoustics and Vibration (2017–present).
- Ranked third for KTH Assistant Professor in Engineering Acoustics (2023).
- Finalist for Assistant Professor in Acoustic Materials, Eindhoven University of Technology (2022).
- Marie Skłodowska-Curie Postdoctoral Fellowship proposal evaluation: 84.6% (2022).
- China Postdoctoral Science Foundation Fellowship (2022); China Scholarship Council Scholarship (2014); Excellent Master Thesis Award (2014); Excellent Graduate Award (2011).

RESEARCH METHODS

Physical and computational acoustics: finite- and boundary-element modelling, vibroacoustics, wave propagation, transfer-matrix methods, optimisation, and machine learning. **Experiment:** acoustic testing and characterisation, Raman spectroscopy, and nanoscale material characterisation. **Programming and data:** MATLAB, Python, R, scientific computing, and statistical analysis.

PUBLICATIONS

BIBLIOMETRIC SUMMARY

13 journal articles · 6 conference papers · 3 granted patents · 3 open-source software packages · 1 invited popular-science article · 475 citations · h-index 9 · i10-index 9. Google Scholar metrics updated 1 October 2026.

Preprint

Liu, Z., Oberman, T., Fang, X., Indolia, N., Aletta, F., Habibovic, A., Kang, J. (2025). "The Influence of Visual Context on Truck Cabin Soundscape Perception." *TechRxiv*. doi:10.36227/techrxiv.176366537.73500767/v1.

Journal articles

Hao, Y., Liu, D., Luo, L., Mu, J., Wang, H., **Liu, Z.**, Li, J., Zhu, Z., Guo, Q., Yang, B. (2025). "Searching for topological semi-complete bandgap in elastic truss lattices." *Advanced Science*. doi:10.1002/advs.202511884.

Xia, Y., Chen, Y., **Liu, Z.**, Guo, Q. (2026). "Two-dimensional chiral Weyl points in an interlaced Kagome layer." *Optics Letters*, 51, 2888–2891. doi:10.1364/OL.599416.

Dai, H., Wang, Y., **Liu, Z.**, Liu, Y., Guo, Y., Liu, D. (2025). "Interlayer phononic energy dissipation in the friction of graphene layers." *Science Advances*, 11, eadu2880.

Dai, H., Wang, Y., Zhao, J., Liu, H., **Liu, Z.**, Liu, D. (2024). "Enhanced double resonance Raman scattering in multilayer graphene with broadband coherent anti-Stokes Raman spectroscopy." *Nanoscale*, 16, 1247–1253.

Wang, Y., Dai, H., **Liu, Z.**, Liu, D. (2023). "Phonon scattering in monolayer molybdenum disulfide under different defect concentrations based on temperature-dependent Raman spectra." *Journal of Physical Chemistry C*, 127, 1109–1116.

Wang, Y., Dai, H., **Liu, Z.**, Liu, D. (2023). "Van Hove singularity modulation of phonon transport in twisted bilayer graphene." *Journal of Physical Chemistry C*, 127, 23752–23759.

Xie, C., Meng, L., **Liu, Z.**, Yang, F., Jiang, X., Yang, J. (2023). "Multifunctional integrated underwater sound absorption materials: a review." *Applied Sciences*, 13, 5638.

Li, X., Zhang, R., **Liu, Z.**, Pu, Y. (2022). "Molecular dynamics study on friction of the iron-aluminum alloy." *Materials Today Communications*, 33, 104402.

Liu, Z., Rumpler, R., Sun, H., Li, Q., Liu, D., Yu, W. (2022). "Improving sound insulation near ring and coincidence frequencies of cylindrical sandwich shells." *International Journal of Mechanical Sciences*, 235, 107661.

Liu, Z., Rumpler, R., Feng, L. (2021). "Locally resonant metamaterial curved double wall to improve sound insulation at the ring frequency and mass-spring-mass resonance." *Mechanical Systems and Signal Processing*, 149, 107179.

Liu, Z., Rumpler, R., Feng, L. (2019). "Investigation of sound transmission through a locally resonant metamaterial cylindrical shell in the ring-frequency region." *Journal of Applied Physics*, 125, 115105.

Song, Y., Feng, L., **Liu, Z.**, Wen, J., Yu, D. (2019). "Suppression of vibration and sound radiation of a sandwich plate via periodic design." *International Journal of Mechanical Sciences*, 150, 744–754.

Liu, Z., Rumpler, R., Feng, L. (2018). "Broadband locally resonant metamaterial sandwich plate for improved noise insulation in the coincidence region." *Composite Structures*, 200, 165–172.

Conference proceedings

- Liu, Z.**, Yu, W. (2026). "Acoustic properties of recycled EPS granular media and their effect on the sound absorption of Helmholtz resonators." *Proceedings of ICSV 32*, Istanbul.
- Liu, Z.**, Fang, X., Oberman, T., Aletta, F., Bolin, K., Habibovic, A., Kang, J. (2026). "Natural sounds in truck cabins: a soundscape modulation study." *Proceedings of ICSV 32*, Istanbul.
- Liu, Z.**, Fang, X., Nicola, N., Oberman, T., Aletta, F., Kang, J. (2024). "An engineering framework towards a better soundscape in truck cabins using metamaterial absorber." *Proceedings of ICSV 30*, Amsterdam.
- Liu, Z.**, et al. (2021). "Design of curved sandwich panel to overcome ring-frequency and coincidence effects." *Proceedings of ICSV 27*.
- Liu, Z.**, Feng, L., Rumpler, R. (2018). "Acoustic behaviour of a locally resonant metamaterial curved panel." *Proceedings of ICSV 25*, 3409–3416.
- Liu, Z.**, Feng, L., Rumpler, R. (2017). "Design of broadband acoustic metamaterials for low-frequency noise insulation." *Journal of the Acoustical Society of America*, 141, 3574.

Open-source research software

- **MetaSandwich-IL-Jacket** (2026): transfer-matrix analysis of pipeline-jacket insertion loss. [GitHub](#).
- **MetaSandwich-STL** (2026): sound-transmission-loss toolbox for acoustic metamaterial panels. [GitHub](#).
- **acoustic-sandwich-waves** (2026): dispersion and wavenumber tools for acoustic wave propagation in sandwich structures. [GitHub](#).

Public engagement and thesis

- Liu, Z.**, Rumpler, R. (2025). "Sonisk Alkemi – Metamaterial och ljudlandskap för en tystare, grönnare stad." *Bygg & teknik*, invited article. [Article](#).
- Liu, Z.** (2019). *Design of Soundproof Panels via Metamaterial Concept*. Doctoral thesis, KTH Royal Institute of Technology. [DiVA](#).